

Name: Javaughn Stevens

Date: Sunday, December 28, 2025

MATLAB 1

% For all plots, the horizontal (x) axis represents t and the vertical (y) axis represents $y(t)$.

Exercise 1 (B. Hunt)

Consider the initial value problem: $ty' + 3y = 5t^2$, $y(2) = 5$

a. Solve using the MATLAB function **dsolve**

Command Window Screenshot:

```
Command Window
>> S = dsolve('t*Dy + 3*y = -5*t^2', 'y(2) = 5', 't')
Warning: Support for character vector or string inputs will be removed in a future release. Instead, use syms to declare variables and replace inputs such as
dsolve('Dy = -3*y') with syms y(t); dsolve(diff(y,t) == -3*y).
> In dsolve (line 126)

S =

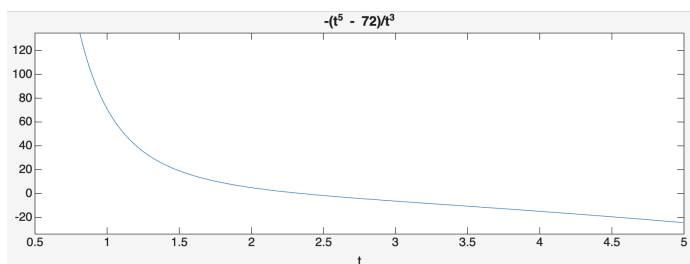
-(t^5 - 72)/t^3

>>
```

% Please note, this warning is only present because I am unable to download a newer version of MATLAB at this time.

b. Plot the solution $y(t)$ on the intervals $0.5 \leq t \leq 5$ and $0.2 \leq t \leq 20$. Determine the behavior of $y(t)$ as t approaches 0 from the right and as t becomes large. Copy your commands in the command window, and the plot, just like above, to the word file

Plot #1: (interval from $0.5 \leq t \leq 5$)



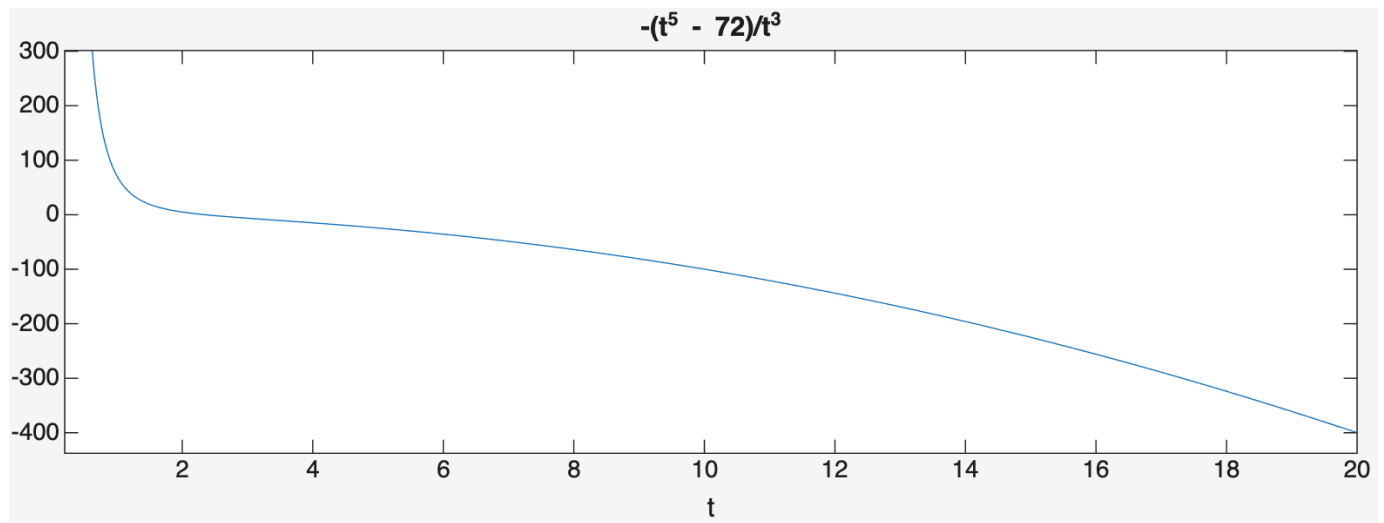
Command Window Screenshot #1:

Command Window

```
>> yt = dsolve('t*Dy + 3*y = -5*t^2', 'y(2)=5', 't');  
ezplot(yt, [0.5, 5])  
Warning: Support for character vector or string inputs will be removed in a future release. Instead, use syms to declare variables and replace inputs such as  
dsolve('Dy = -3*y') with syms y(t); dsolve(diff(y,t) == -3*y).  
> In dsolve (line 126)
```

As $t \rightarrow 0^+$, $y(t) \rightarrow \infty$, and as $t \rightarrow \infty$, $y(t) \rightarrow -\infty$.

Plot #2: (interval from $0.2 \leq t \leq 20$)



Command Window Screenshot #2:

Command Window

```
>> yt = dsolve('t*Dy + 3*y = -5*t^2', 'y(2)=5', 't');  
ezplot(yt, [0.2, 20])  
Warning: Support for character vector or string inputs will be removed in a future release. Instead, use syms to declare variables and replace inputs such as  
dsolve('Dy = -3*y') with syms y(t); dsolve(diff(y,t) == -3*y).  
> In dsolve (line 126)  
>>
```

As $t \rightarrow 0^+$, $y(t) \rightarrow \infty$, and as $t \rightarrow \infty$, $y(t) \rightarrow -\infty$.

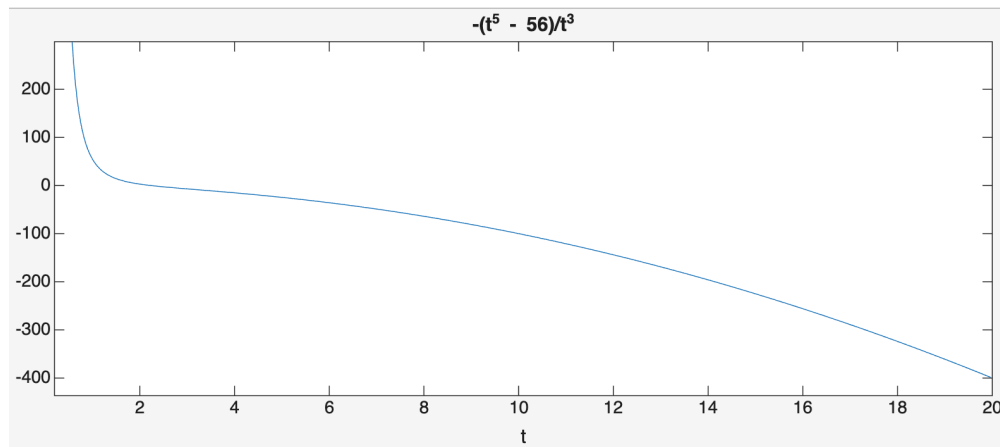
c. Change the initial condition to $y(2) = 3$ and determine the behaviour of this

solution, again by plotting on intervals such as $0.2 \leq t \leq 20$ and $0.5 \leq t \leq 5$. Copy

your commands in the command window, and the plot, just like above, to the word

file.

Plot 1: (interval from $0.2 \leq t \leq 20$)



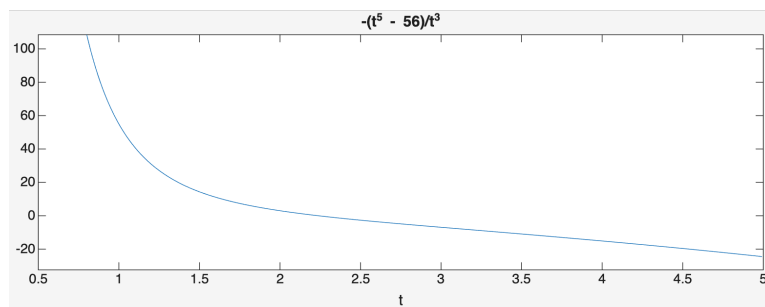
Command Window Screenshot #1:

```
Command Window
>> yt = dsolve('t*Dy + 3*y = -5*t^2', 'y(2)=3', 't')
ezplot(yt, [0.2, 20])
Warning: Support for character vector or string inputs will be removed in a future release. Instead, use syms to declare variables and replace inputs such as
dsolve('Dy = -3*y') with syms y(t); dsolve(diff(y,t) == -3*y).
> In dsolve (line 126)

yt =
-(t^5 - 56)/t^3
```

As $t \rightarrow 0^+$, $y(t) \rightarrow \infty$, and as $t \rightarrow \infty$, $y(t) \rightarrow -\infty$.

Plot #2: (interval from $0.5 \leq t \leq 5$)



As $t \rightarrow 0^+$, $y(t) \rightarrow \infty$, and as $t \rightarrow \infty$, $y(t) \rightarrow -\infty$.

Command Window Screenshot #2:

```

Command Window
>> yt = dsolve('t*Dy + 3*y = -5*t^2', 'y(2)=3', 't')
ezplot(yt, [0.5, 5])
Warning: Support for character vector or string inputs will be removed in a future release. Instead, use syms to declare variables and replace inputs such as
dsolve('Dy = -3*y') with syms y(t); dsolve(diff(y,t) == -3*y).
> In dsolve (line 126)

yt =

-(t^5 - 56)/t^3

```

Exercise 2 (B. Hunt)

a. Solve the initial value problem: $y' - 2y = \sin(2t)$, $y(0) = c$

We can rewrite this IVP in terms of dy/dt : $y' = dy/dt = \sin(2t) + 2y$, $y(0) = c$.

Command Window Screenshot:

```

Command Window
>> S = dsolve('Dy = sin(2*t) + 2*y','y(0) = c', 't')
Warning: Support for character vector or string inputs will be removed in a future release. Instead, use syms to declare variables and replace inputs such as
dsolve('Dy = -3*y') with syms y(t); dsolve(diff(y,t) == -3*y).
> In dsolve (line 126)

S =

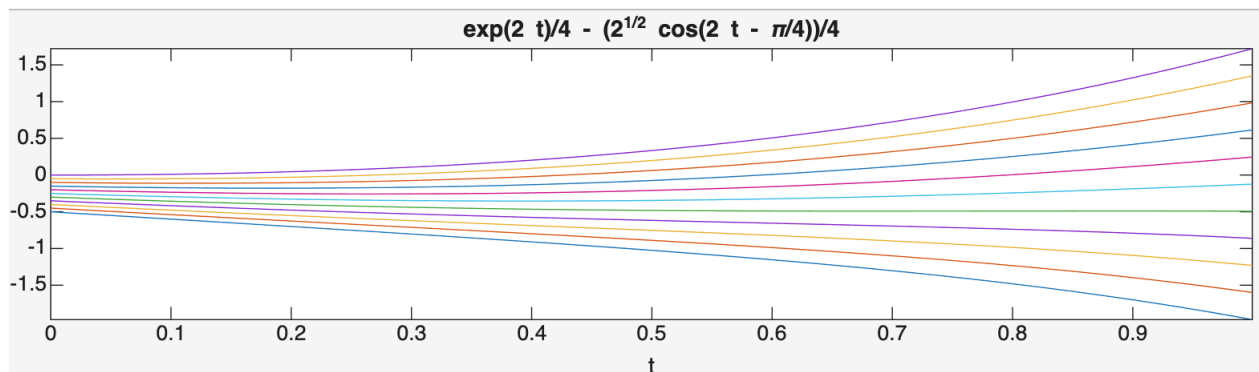
exp(2*t)*(c + 1/4) - (2^(1/2)*cos(2*t - pi/4))/4

```

b. Use MATLAB to graph solutions for $c = -0.5, -0.45, \dots, -0.05, 0$ on the interval

$[0, 1]$. Use axis tight.

Graph:



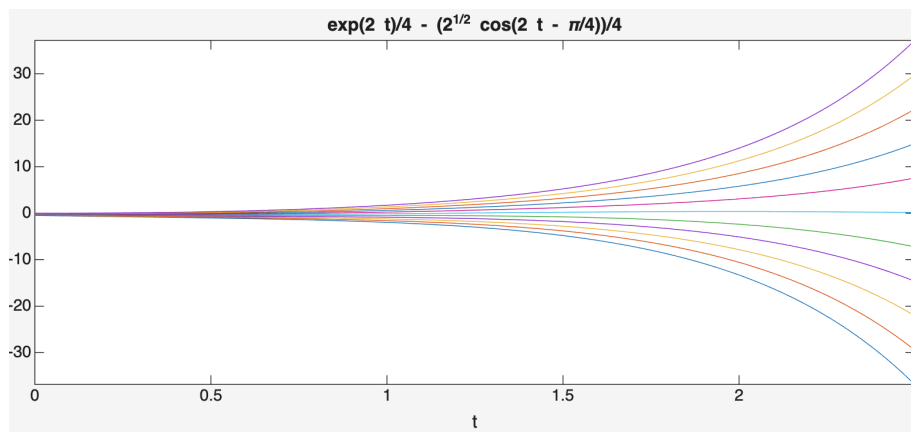
Command Window Screenshot:

Command Window

```
>> S = dsolve('Dy = sin(2*t) + 2*y', 'y(0) = c', 't')
for cs = -0.5: 0.05: 0
    Scs = subs(S, 'c', cs);
    ezplot(Scs, [0,1])
    hold on
end
hold off
axis tight;
```

c. Use MATLAB to graph solutions for $c = -0.5, -0.45, \dots, -0.05, 0$ on the interval $[0, 2.5]$

Graph:



Command Window Screenshot:

Command Window

```
>> S = dsolve('Dy = sin(2*t) + 2*y', 'y(0) = c', 't')
for cs = -0.5: 0.05: 0
    Scs = subs(S, 'c', cs);
    ezplot(Scs, [0,2.5])
    hold on
end
hold off
axis tight;
Warning: Support for character vector or string inputs will be removed in a future release. Instead, use syms to declare variables and replace inputs such as
dsolve('Dy = -3*y') with syms y(t); dsolve(diff(y,t) == -3*y).
> In dsolve (line 126)

S =
exp(2*t)*(c + 1/4) - (2^(1/2)*cos(2*t - pi/4))/4
```

d. What happens to the solution curves as t increases? Describe in words.

As t increases, the solution curves begin to spread out, with the top and bottom curves blowing up and down respectively (changing without bounds).

e. What effect can small changes in initial data have on the global behavior of solutions curves?

Small changes in initial data can produce both small or significant changes on the global solution curves, depending on stability. More stable solutions will change less compared to more unstable solutions.

Note you can label axes:

```
xlabel('t')
```

```
ylabel('y')
```